

AI-Driven Customer Churn Prevention System

Telecom Company A — Data Science Consulting Project



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Business Problem

Customer churn leads to significant and continuous revenue loss in Telecom Company A.

From historical customer data, nearly 50% of customers are classified as churned, indicating a critical retention issue.

Key challenges:

- *Lack of early identification of at-risk customers*
- *Limited understanding of churn drivers*
- *Inefficient retention targeting strategies*

Therefore, Company A requires a data-driven churn prediction system to proactively identify and retain high-risk customers.



Project Objective

- *Analyze customer behavioral and billing data to understand churn drivers*
- *Build a machine learning model to predict customer churn*
- *Identify high-risk customers early*
- *Provide actionable retention strategies to reduce revenue loss*



Dataset Description

- ***100,000 customers***
- ***Two datasets:***
 - ***Customer profile (demographics, device info)***
 - ***Usage history (calls, revenue, behavior)***
- ***Target variable: churn (binary classification)***
- ***Balanced dataset (~50% churn / ~50% non-churn)***



Data Approach

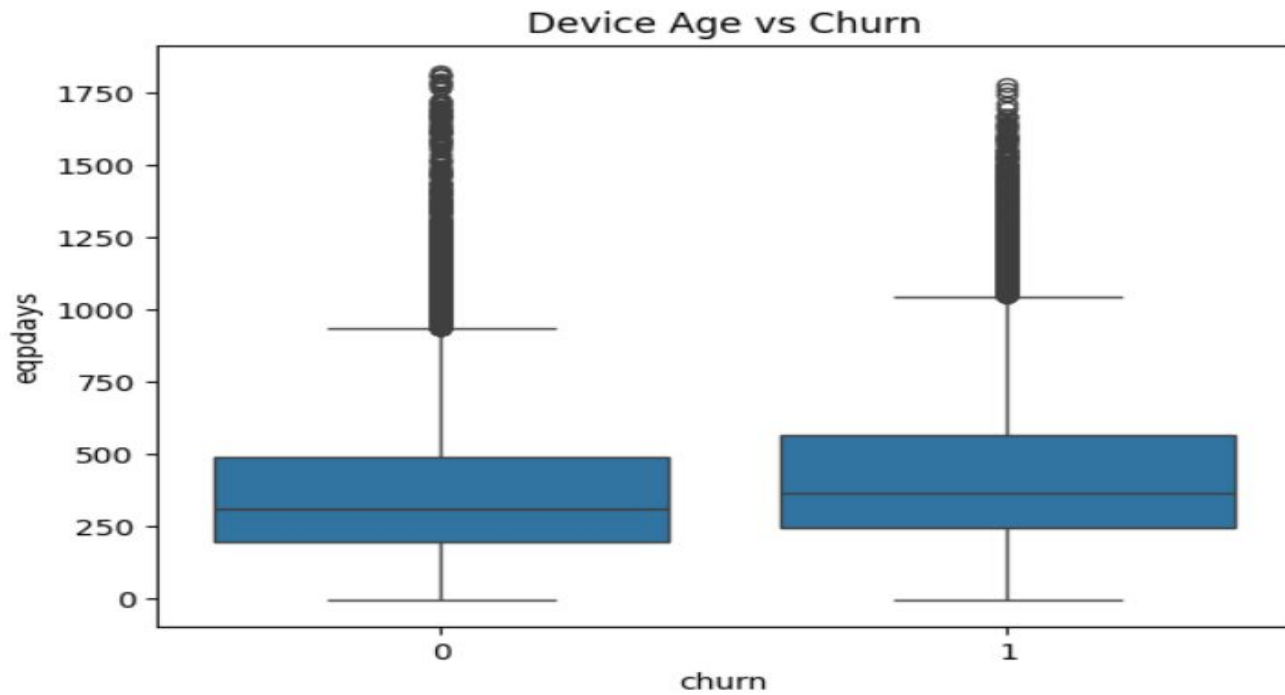
Methodology

- 1. Data Cleaning & Exploration**
- 2. Exploratory Data Analysis (EDA)**
- 3. Correlation Analysis**
- 4. Machine Learning Models**
 - **Logistic Regression**
 - **Random Forest**
- 5. Model Evaluation (Accuracy, ROC-AUC)**



Key Insight 1 (Device Age)

- Customers with older devices (higher eqpdays) are more likely to churn
- Device age is positively associated with churn risk
- This suggests that handset replacement/upgrading can improve retention

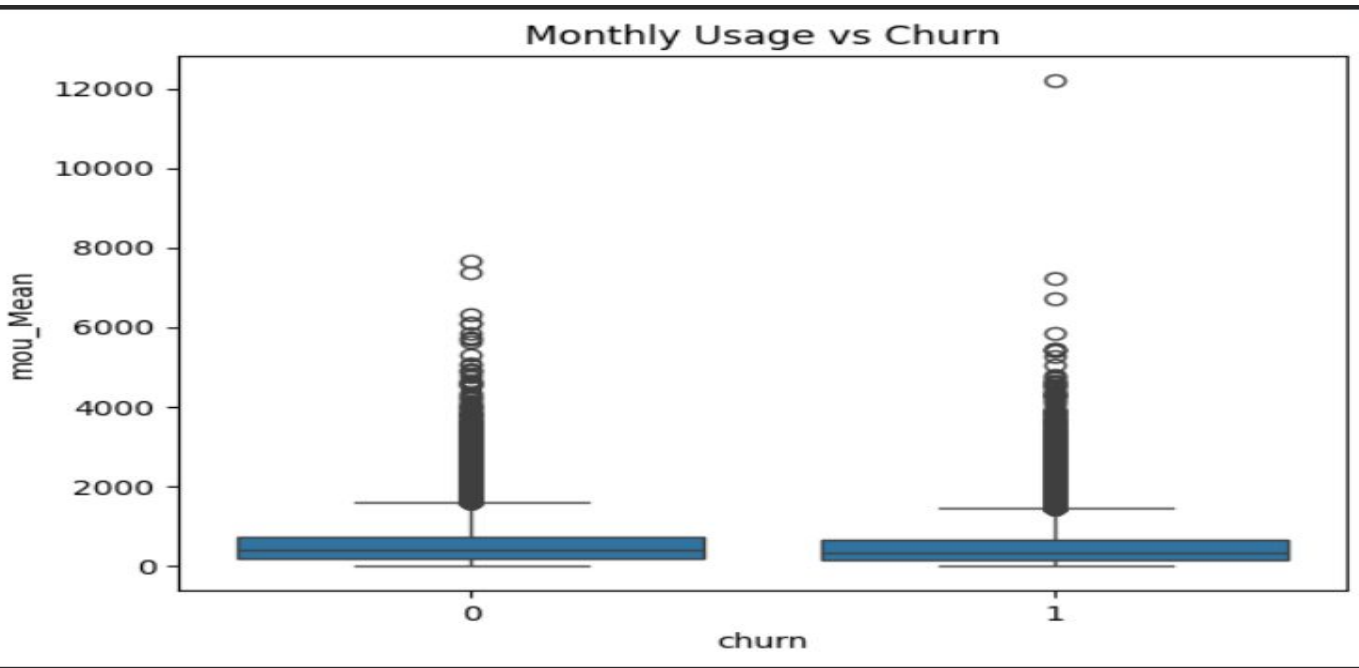




Key Insight 2 (Usage Behavior)

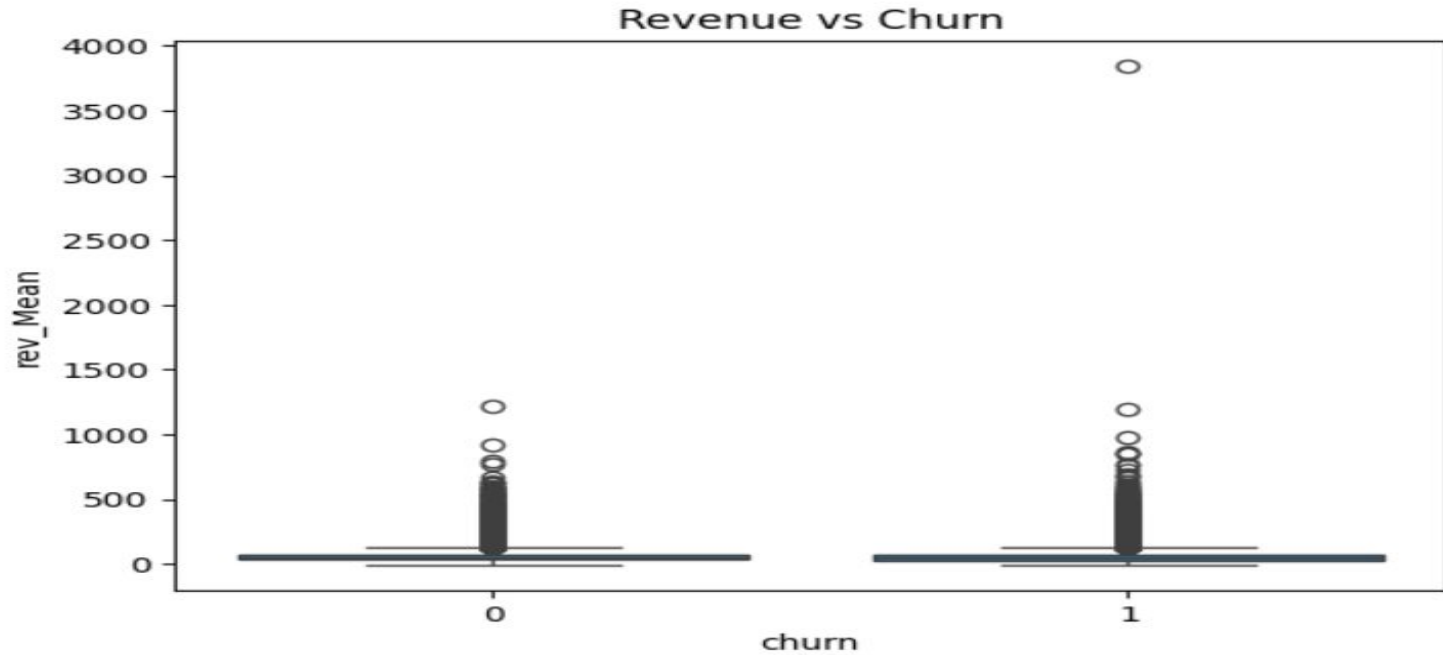
nsight: Usage intensity affects churn

- Low usage customers have higher churn probability
- High engagement → higher retention



Customers with irregular or declining revenue are at risk
Overages and revenue drops correlate with churn

Overages and revenue drops correlate with churn





Correlation Summary

Positive correlation:

- *eqpdays (device age)*
- *change_mou (usage change)*

Negative correlation:

- *usage features (mou, calls)*
- *revenue stability*

👉 Conclusion:

Churn is driven by behavioral + device-related factors



Modeling Approach

Machine Learning Models

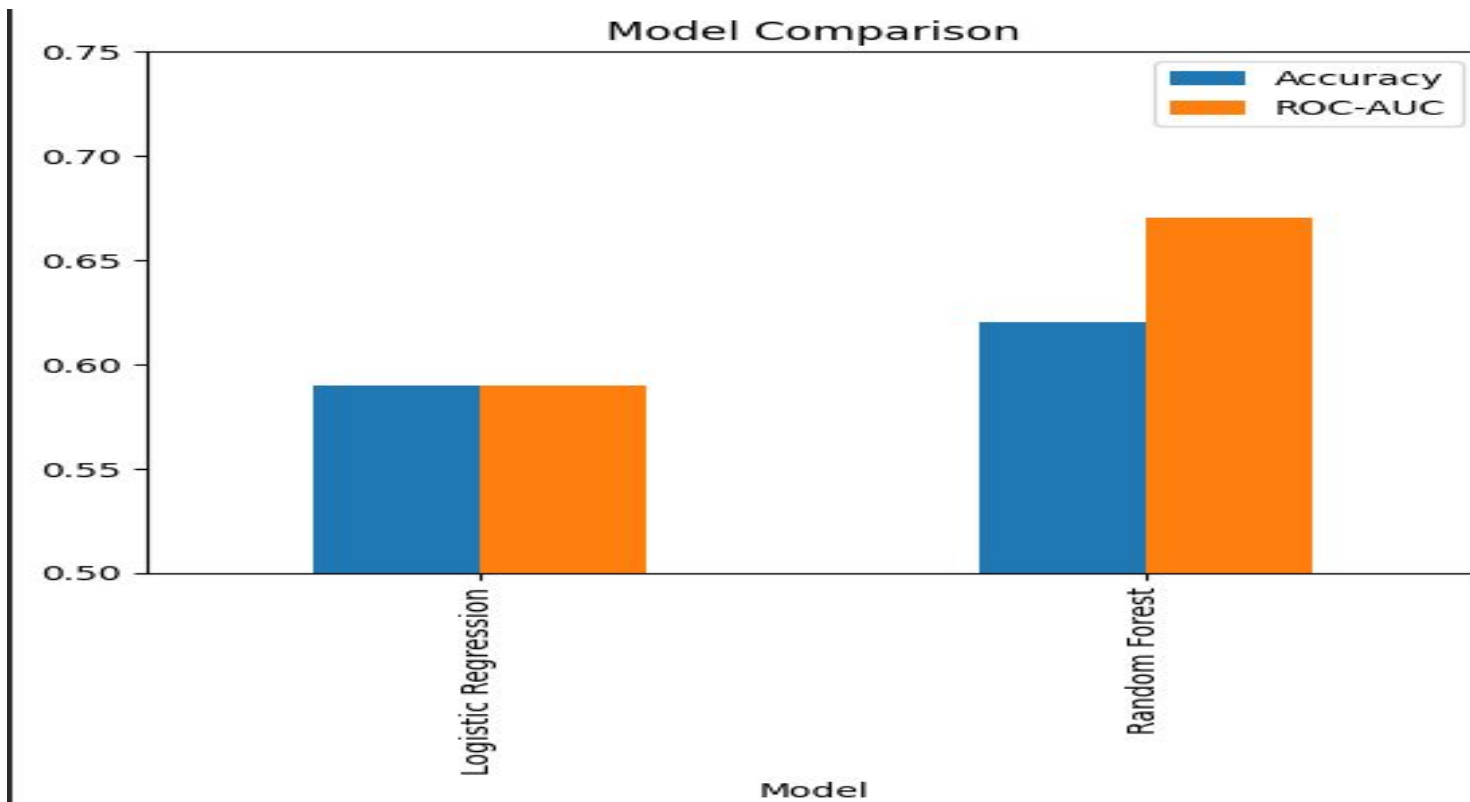
- *Logistic Regression (baseline)*
- *Random Forest (final model)*

Why:

- *Logistic → interpretability*
- *Random Forest → handles non-linear patterns*



Model Performance

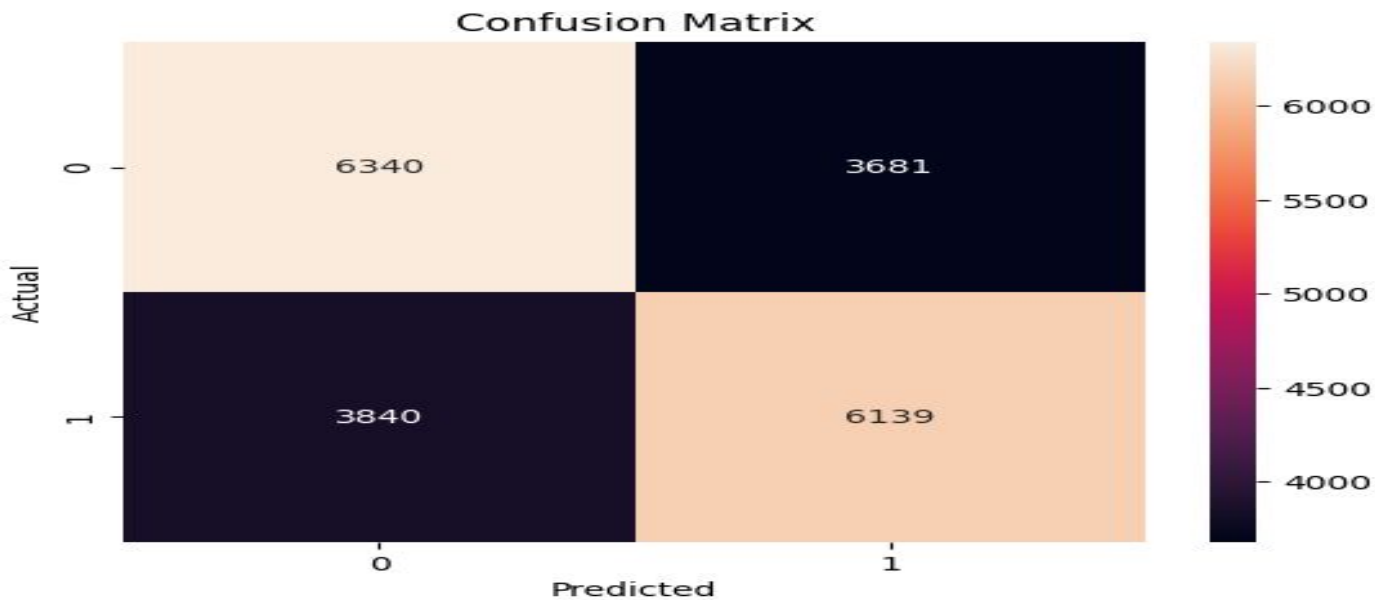




Confusion Matrix

Balanced performance across classes

Model successfully captures churn patterns better than random baseline





Business Solution

Proposed Solution: AI Churn Prevention System

System includes:

- ***Churn prediction model***
- ***Customer risk scoring***
- ***Segmentation of customers (high / medium / low risk)***
- ***Early warning system for churn risk***



Retention Strategy & Impact

Recommended Actions

High-risk customers:

- *Device upgrade offers*
- *Discounted plans*

Usage drop customers:

- *Personalized bundles*
- *Loyalty incentives*

High-value customers:

- *VIP retention programs*